

Designing Effective Inputs

[Chapter 12]

System Analysis and Design

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- The quality of system input determines the quality of system output
- Input design objectives:
 - Effectiveness—all serve specific purposes for users of the information system.
 - Accuracy—design that ensures proper completion.
 - Ease of use—straightforward and require no extra time to decipher.
 - Consistency—group data similarly from one application to the next.
 - Simplicity—keeping designs uncluttered in a manner that focuses the user's attention.
 - Attractiveness—appealing design.



- Form flow-Design forms with proper flow, from left to right and top to bottom.
- Seven sections of a form-Group information logically using the seven sections of a form.
- Captioning-Provide people with clear captions.



Illogical flow takes extra time and is frustrating.

- Can minimize the time and effort expended by employees in form completion
- Should flow from left to right and top to bottom

Seven Sections of a Form



- Heading
- Identification and access
- Instructions
- Body
- Signature and verification
- Totals
- Comments



Captions tell the person filling out the form what to put in a blank line, space, or box.

- Line caption
 - Putting the caption on the same line or below the line
- Boxed caption
 - Providing a box for data instead of a line
- Check off caption
 - Lining up choices or alternatives vertically
- Horizontal check off caption
 - Lining up choices or alternatives horizontally
- Table caption
 - Work well in the body of a form
- Combination

7



- Make sure that each form in use fulfills its specific purpose
- Make sure that the specified purpose is integral to organizational functioning
- Prevent duplication of information collected and of the forms that collect it
- Design effective forms
- Decide on how to reproduce forms in the most economical way
- Establish procedures that make forms available, at the lowest possible cost



GUI is sometimes referred to as point-and-click.

- Text boxes
- Check boxes
- Option or radio buttons
- List and drop-down list boxes
- Sliders and spin buttons
- Image maps
- Text area
- Message boxes



- Placeholder puts a small amount of help text into a text box, displaying in a lighter color
- When the cursor is placed in the field the text vanishes
- New text boxes:
 - Email
 - Telephone
 - URL, a Web address
- These appear as normal text boxes on a computer



- When used on a tablet or smartphone, they can be used to customize the pop-up keyboard
 - Telephone number, the keyboard layout changes to a number pad
 - A URL, the keyboard includes a .com button
 - An email address, the keyboard includes an @ symbol
- Helps the user enter data quickly and accurately
- A data list displays a drop-down list of predefined suggestions to make entry easier for the user
- As the user begins to type the first few letters, the datalist is displayed



- The user can choose one of the items in the list to make a selection
- This is used in the autocomplete function
- A calendar control to select a date, a date and time, or a local date and time
- Selecting dates from a pop-up calendar is easier and less error-prone than entering text



- Use stacked layers
- Code is moved in front of the web page
- Code may be created using JavaScript
- Analyst must determine when to use layers



- Analyst must determine:
 - How is the layer built?
 - What events cause the layer to be created?
 - What events remove the layer?
 - Where should the block be placed?
 - The size of the block?
 - How to frame the region?
 - What happens when an option is selected?